Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech
Code: 100302

Subject: Analog Electronic Circuits

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven questions only):-*** **[2 x 7 = 14]**

(a) A Zener diode rated 5.1V is used with a 220Ω resistor. If input is 12V, what is the Zener current?
(i) 31.4 mA
(ii) 50 mA
(iii) 20 mA
(iv) 10 mA

(b) At room temperature, a transistor has a collector current of $5\ mA$ and $V_{BE} = 0.7V$. Find g_m
(i) $7.14\ S$
(ii) $194\ mS$
(iii) $5\ S$
(iv) 3.5 S

(c) Which region does a MOSFET operate in for amplification?
(i) Cutoff
(ii) Triode
(iii) Saturation
(iv) Ohmic

(d) The trans-conductance (gm) of a MOSFET is a measure of:
(i) Voltage gain
(ii) Input impedance
(iii) Output conductance
(iv) Change in drain current per unit change in gate voltage

(e) Which parameter limits the maximum frequency response of an op-amp?
(i) Slew rate
(ii) Offset voltage
(iii) Input bias current
(iv) Common-mode gain

(f) An inverting amplifier has Rf = 100 kΩ and Rin = 10 kΩ. What is the voltage gain?
(i) -100
(ii) -10
(iii) 10
(iv) 1

(g) Which op-amp circuit converts a sinusoidal input to a square wave output?
(i) Peak detector
(ii) Comparator
(iii) Schmitt trigger
(iv) Integrator

(h) A MOSFET has $I_D = 2\ mA$ and $V_{GS} = 2.5$. If Vth = 1 V and $k_n = 1$ mA/V², what is the value of I_D.
(i) 1 mA
(ii) 2 mA
(iii) 3.5 mA
(iv) 4 mA

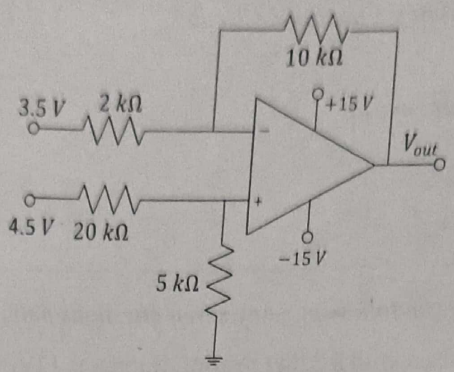
(i) Which component is used to maintain a constant voltage across the load?
(i) P-N junction diode
(ii) Zener diode
(iii) Schottky diode
(iv) Tunnel diode

(j) Which configuration of BJT offers the highest current gain?
(i) Common Base
(ii) Common Collector
(iii) Common Emitter
(iv) Emitter Follower

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Q.2 (a) Explain the working of a differential amplifier using BJTs. Derive expressions for differential and common-mode gain.

(b) Assume the op-amp is ideal, find the output voltage V_{out}.

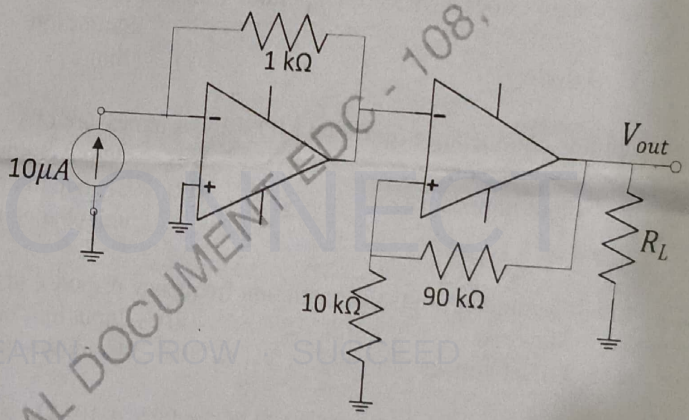


Q.3 (a) Design and analyze a Wein Bridge Oscillator using op-amp. Derive the expression for frequency of oscillation.

(b) Explain the operation of a Zener diode as a voltage regulator. Design a circuit to provide 5V regulated output.

Q.4 (a) Explain the operation of a hysteretic comparator using op-amp. Draw and analyze the transfer characteristics. [7]

(b) Assume the op-amp is ideal, find the output voltage V_{out}. [7]

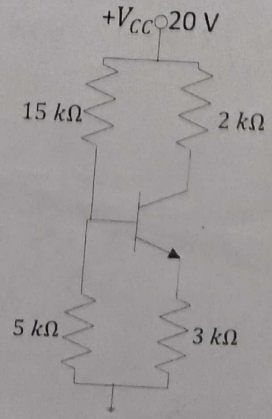


Q.5 (a) Design and analyze a clamping circuit. Explain how it shifts the DC level of an input waveform. [7]

(b) Explain the effect of channel length modulation on the performance of a MOSFET amplifier. [7]

Q.6 (a) Design a triangular wave generator using op-amps. [7]

(b) For the circuit shown below, draw D.C. load line. Assume β is very large. [7]



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Q.7 (a) Design a first-order low-pass butter worth filter circuit using op-amp and derive the expression of high cut off frequency and transfer function of the filter. [7]

(b) Explain the role of op-amp in waveform shaping circuits with examples. [7]

Q.8 (a) Derive the expression for ripple voltage and peak inverse voltage in a full-wave rectifier. [7]

(b) Explain the various biasing techniques for a BJT amplifier. Suggest which one is better and why? [7]

Q.9 Write short notes on ***any two*** of the following:- [7 x 2 = 14]

(a) Current mirror using BJT

(b) Trans-conductance of MOSFET

(c) Junction break-down in Zener diodes

(d) high-frequency equivalent circuit of BJT

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